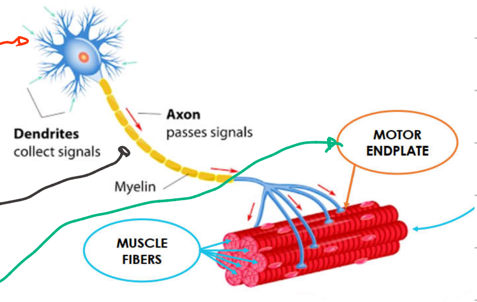
Brief Recap

Motor Unit : Motor neuron + Muscle Fiber

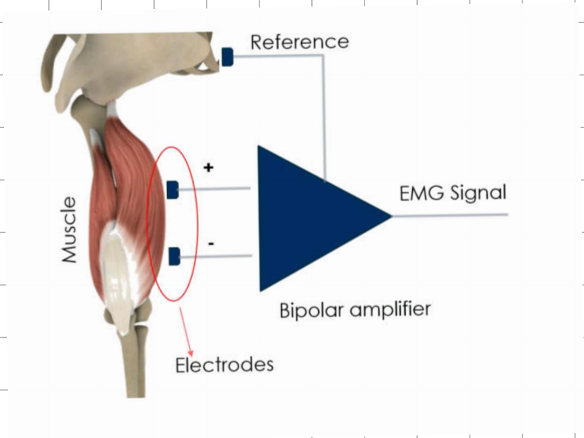


Basically a signal is generated at the level of the spinal cord
↳ it travels through the axon
↳ And then innervates the muscular Fibers, through the motor endplate

Action Potential: nerve impulse reaches the motor endplates causing the contraction.

Motor Unit Action Potential: sum of APs of a single motor unit.

Motor Unit Action Potential Train (MUAPT): Train of MUAPs.



How can we record this signal?

We use some electrodes able to convert the signal generated by the ions inside our body to an electrical signal

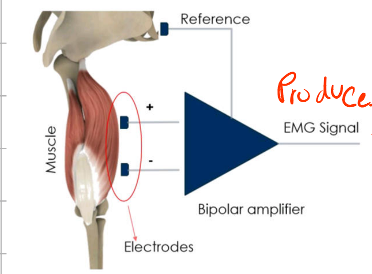
We prefer Bipolar Configuration (Diff amplifier) over monopolar

The two electrodes are placed on the active volume, on the muscle we wanna register + one electrode placed far away as reference. This provides high → Resolution and CMRR

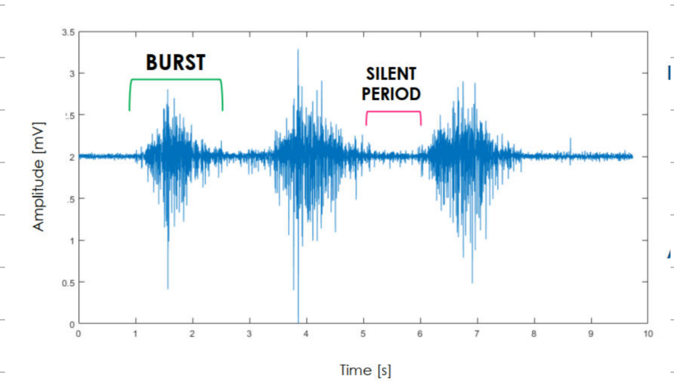
Electrodes convert the action potential generated by the ions in electric signal.

Differential amplifier uses two electrodes in the active volume and one as reference. unit.

High resolution and common mode noise rejection.



Produces then →



Burst period → Muscle contraction
Silent period → Resting phase of the muscle

EMG signal is an stochastic noisy signal and we need to process it to extract useful information

Amplitude

- 0.5 – 5 mV → 5mV in the case of big muscles

EMG Bandwith

- Depth electrodes (neurophysiology) 10 Hz – 10 kHz
- Surface electrodes (kinesiology) 10 Hz – 500 Hz

→ Useful information can be found on the envelope

in order to do this, there is a pipeline

1. We acquire the raw signal
2. We perform the Filtering of the BW $\boxed{10-500\,Hz}$

What do we Filter? → The exogenous and endogenous noise

Power Supply (50–60 Hz)
Skin – Electrode Interface (0–10 Hz)
Movement Artifacts (0–10 Hz)

Other muscles or Heart (1 Hz)

Power Supply (50-60 Hz)
Skin - Electrode Interface (0-10 Hz)
Movement Artifacts (0-10 Hz)

Other muscles or
Heart (1 Hz)

Filtering
- Notch Filter (50 Hz)
- HPF $f_c = 10$ Hz

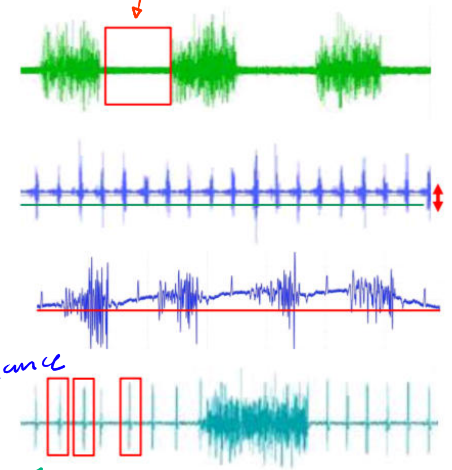
NOISE EFFECTS

- Power supply (50 – 60 Hz)
- Baseline offset (DC component)
- Baseline drift (e.g. Movement artifacts)

Due to change in Electrode-Electrolyte impedance

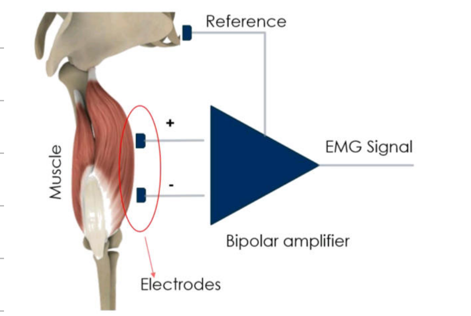
- ECG artifacts (1 Hz)

non-zero voltage at baseline level



Very high amplitude

and sometimes Filtering is not enough but Electrodes must be placed as far as possible from the heart



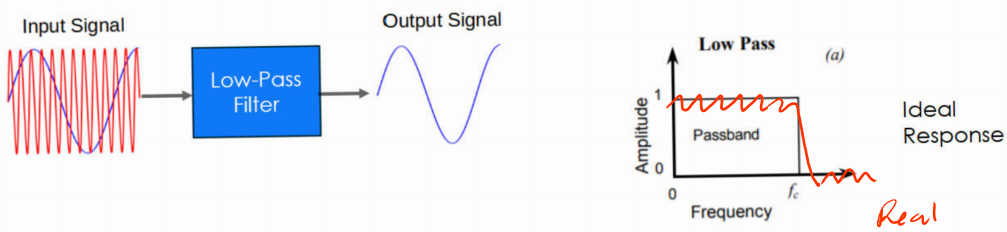
Increase the SNR

Antialiasing

$f_s \geq 2 \cdot f_{max}$ For ADC

So, let's start.

Low pass filter:



Ideal Response

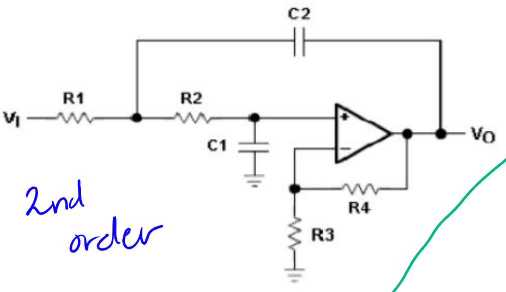
Real Response
Ripples & Transition band

$$\frac{\text{Vout}}{\text{Vin}} = \frac{1}{sCR+1}$$

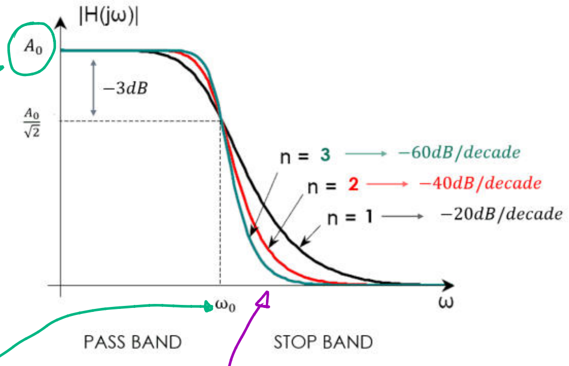
$$\text{fc} = \frac{1}{2\pi RC}$$

To avoid the ripples in the pass-band we implement a Butterworth (which still contains the transition band)

Low pass filter: Butterworth



2nd order



$$|H(\omega)| = \frac{A_0}{\sqrt{1+\left(\frac{\omega}{\omega_0}\right)^{2n}}}$$

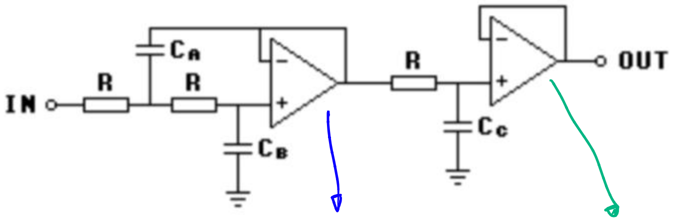
ω_0 = cut-off pulsation
ω = input signal pulsation
A_0 = max gain in the band pass
n = order of the filter

Flat response in the pass-band and the larger the order, the sharper the transition band

→ We aim for this if we wanna preserve amplitude linearity of a signal

No matter what filter, there is always an attenuation at the level of the cutoff freq ω_0 of $-3dB$

Low pass filter: III order Butterworth

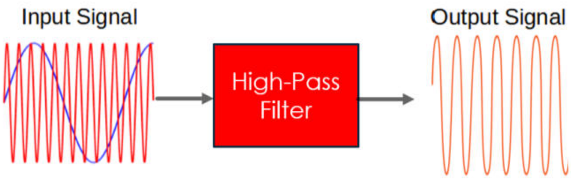


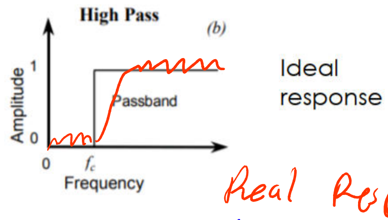
Cascade of 2nd BW LPF + 1st BW LPF

n	Butterworth Polynomials [$\omega_n = 1$]:
1	$(s+1)$
2	$(s^2 + s\sqrt{2} + 1)$
3	$(s+1)(s^2+s+1)$
4	$(s^2 + 0.765s + 1)(s^2 + 1.848s + 1)$

Let's move on to the HPF

High pass filter:



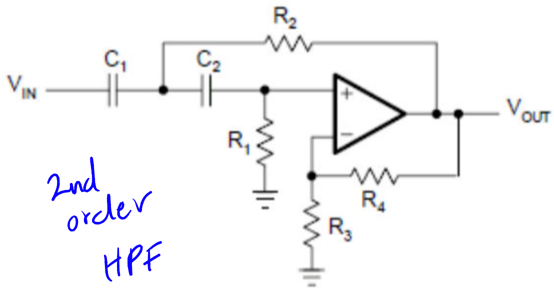


Real Response.
Transition band.

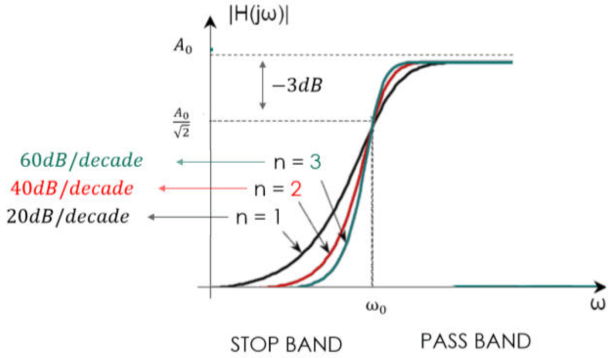
$$\boxed{\frac{Vout}{Vin} = \frac{sCR}{sCR+1}} \qquad \boxed{fc = \frac{1}{2\pi RC}}$$

We use same approach as the LPF for HPF

High pass filter: Butterworth



2nd order HPF



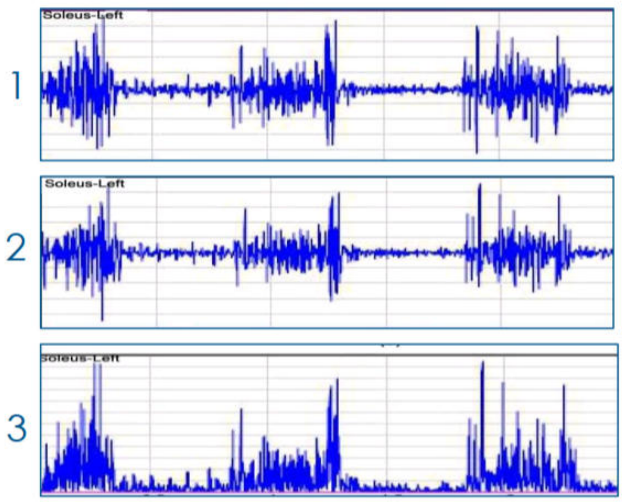
$$\boxed{|H(\omega)| = \frac{A_0}{\sqrt{1+\left(\frac{\omega_0}{\omega}\right)^{2n}}}}$$

ω_0 = cut-off pulsation
ω = input signal pulsation
A_0 = max gain in the band pass
n = order of the filter

which is basically the same configuration but with the capacitors and resistors swapped

Coming back to the envelope extraction.

1. Raw Signal
2. Filtering
3. Rectification

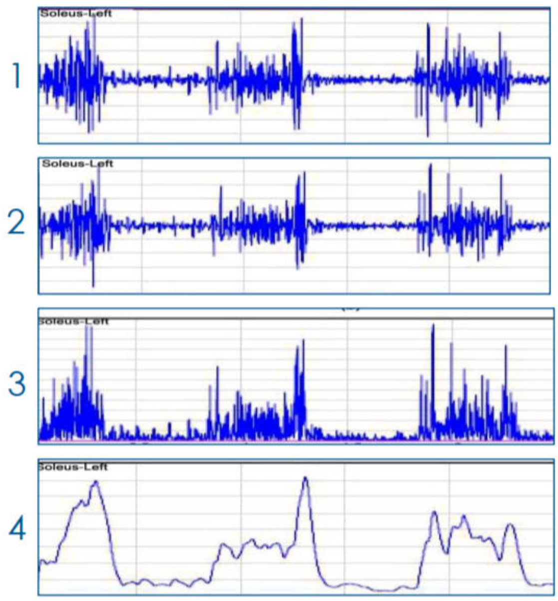


Since EMG is an stochastic noise signal, its mean is $\mu \sim 0$. So full wave or half wave rectification is needed

And now, after performing rectification we can proceed with the extraction of the envelope by means of diff methods

(1) Raw signal
(2) Filtering
(3) FWR

(4) Envelope Extraction



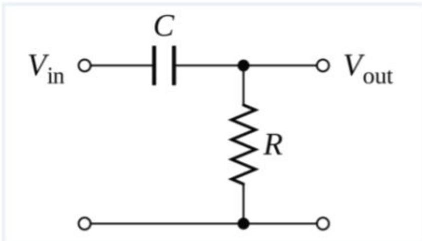
4. Envelope extraction
   a. Moving Average (LP filter) $y[i] = \frac{1}{M}\sum_{j=0}^{M-1} x[i+j]$
      - Time window: 50 – 100 ms
      - Cut-off frequency: 5 – 10 Hz
   b. Root mean square (RMS) $x_{RMS} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}|x_n|^2}$
      - FWR no needed
      - Time window: 50 – 100 ms

Rectification is implicit with the squared in the formula

Exercises !!!

EXERCISE 1 – Exam Jan 1° 2018, exercise 5

Dimension a 1st order high pass filter to eliminate the low-frequency components of the EMG signal.



So, low frequency components are due to
- Skin – Electrode Interface (0–10 Hz)
- Moving Artifacts (0 – 10 Hz)

So we apply a HPF with a $f_c = 10\,Hz$

$f_c = \frac{1}{2\pi RC} = 10\,Hz$ ← we choose R & C

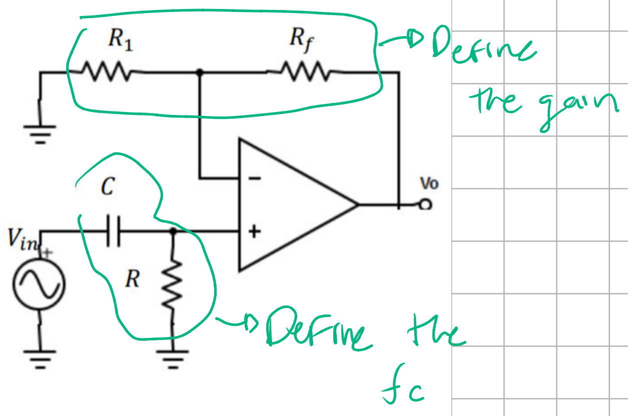
if $R = 10\,k\Omega$

$C = 1.6\,\mu F$

input impedance of following stage needed to be high
- Too small (Ω) → previous stage will be loaded
- Too large (MΩ) → parasitic capacitance
  C will be too small and f_c can be influenced by $C + C_s$

EXERCISE 2 – Exam Sep 3° 2020, exercise 2

Design a 1st order high-pass butterworth filter with 2Hz cut-off frequency by using capacitance of 10 [μF]. What is the required resistance R?



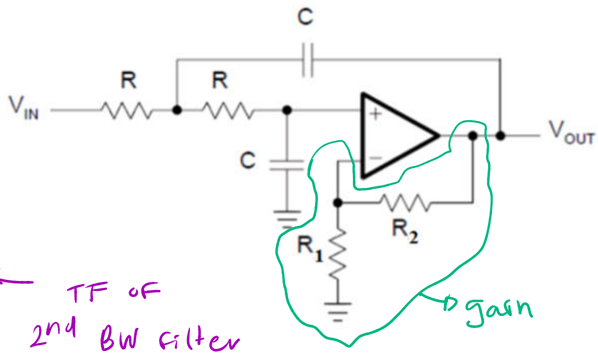
$C = 10\,\mu F$ and $f_c = \frac{1}{2\pi RC} = 2Hz$

Thus $\boxed{R \sim 8K\Omega}$

$7957\,\Omega$

EXERCISE 3

Design a second-order butterworth filter with a cut-off frequency equal to 50Hz.



We start by knowing
$f_c = 50Hz$

$$\frac{V_{out}}{V_{in}} = \frac{A_{v_0}\, W_n}{s^2 + 2\xi W_n s + W_n^2}$$ ← TF of 2nd BW filter

$W = \frac{1}{RC}$ and $\xi = \frac{3 - A_{v_0}}{2}$

A_{v_0} is the gain $A_v = \left(1 + \frac{R_2}{R_1}\right)$

$\xi = \frac{3 - A_{v_0}}{2}$ and remember we can also get the function of the BW filter by using its normalized polynomial

$$s^2 + \underbrace{s\sqrt{2}}_{2\xi W_n} + \underbrace{1}_{W_n^2}$$

$2\xi = \sqrt{2}$

$\xi = \frac{\sqrt{2}}{2}$

$\frac{\sqrt{2}}{\cancel{2}} = \frac{3}{\cancel{2}} - \frac{A_v}{\cancel{2}}$

$A_v = 3 - \sqrt{2} \rightarrow 1 + \frac{R_2}{R_1} = 3 - \sqrt{2}$

$$\frac{R_2}{R_1} = 2-\sqrt{2} \rightarrow \frac{R_2}{R_1} \simeq 0.6$$

$R_2 = 6K\Omega$ and $R_1 = 10\,K\Omega$

Now let's find RC

$$\omega_n = 2\pi \cdot f_c = \frac{1}{RC} \rightarrow 1 = 2\pi \cdot 50Hz \cdot R \cdot C$$

let's set it to

$$\boxed{R = 10K\Omega \quad \text{then} \quad C = 0.32\,\mu F}$$

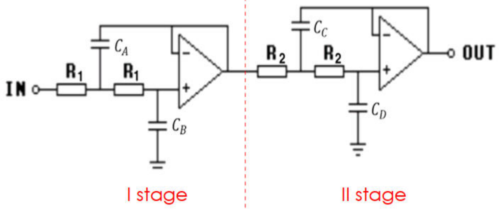
EXERCISE 4

To sense the heart rate variation, a force platform is used. The subject is asked to lie prone on the platform. As expected, the signal recorded as output is very noisy and a filtering stage is required, consisting of a 4th-order butterworth filter with a cut-off frequency equal to 4Hz. Compute R_1, R_2, C_B, C_D components by using:

- $C_A = 6\ \mu F$
- $C_C = 2\mu F$

Use the following known Butterworth polynomial:

$$\boxed{(s^2+0.765s+1)(s^2+1.848s+1)}$$



obtained with this formula

$$B_n(s) = \prod_{k=1}^{n/2}\left[s^2 - 2s\cos\left(\frac{2k+n-1}{2n}\pi\right)+1\right]$$

$$-2\cos\left(\frac{2\cdot 1+4-1}{2\cdot 4}\pi\right) = 0.765 \rightarrow 2\xi_1\omega_{n_1}$$

$$-2\cos\left(\frac{2\cdot 2+4-1}{2\cdot 4}\pi\right) = 1.848 \rightarrow 2\xi_2\omega_{n_2}$$

Let's work with the first stage

$$\omega_{n_1}^2 = \frac{1}{R_1^2 C_A C_B} \quad \therefore \quad \omega_{n_1} = \frac{1}{R_1\sqrt{C_A}\sqrt{C_B}}$$

$$\omega_{n_1} = 2\pi f_c = \frac{1}{R_1\sqrt{C_A C_B}}$$

But also $0.765 \longrightarrow 2\xi_1 \omega_{n_1}$

$$\xi_1 \omega_{n_1} = \frac{1}{R_1 C_A} \rightarrow \xi_1 = \frac{1}{R_1 C_A} \cdot \frac{1}{\omega_{n_1}}$$

$$\xi_1 = \frac{1}{\cancel{R_1} C_A} \cdot \cancel{R_1}\sqrt{C_A C_B} \qquad \xi_1 = \sqrt{\frac{C_B}{C_A}}$$

So the result for our fourth-order filter TF

$$\frac{V_{out}}{V_{in}} = \frac{\frac{1}{R_1\sqrt{C_A C_B}}}{s^2 + \frac{2}{R_1 C_A}s + \frac{1}{R_1^2 C_A C_B}} \cdot \frac{\frac{1}{R_2\sqrt{C_C C_D}}}{s^2 + \frac{2}{R_2 C_C}s + \frac{1}{R_2^2 C_C C_D}}$$

Now, for the first stage, since

$$2\cdot\xi_1 \omega_{n_1} = 0.765 \quad \text{and} \quad \omega_{n_1}$$

$$\xi_1 = \frac{0.765}{2} = \sqrt{\frac{C_B}{C_A \rightarrow 6\mu F}}$$

then $\boxed{C_B = 0.87\,\mu F}$

$$\xi_2 = \frac{1.848}{2} = \sqrt{\frac{C_D}{C_C \rightarrow 2\mu F}}$$

$\boxed{C_D = 1.70\,\mu F}$

$$\underset{2\pi\cdot 4\,Hz}{\omega_{n_1}} = \frac{1}{R_1\sqrt{\underset{6\mu F}{C_A}\,\underset{0.87\mu F}{C_B}}}$$

$\boxed{R_1 = 17.4\,K\Omega}$

$$\underset{2\pi\cdot 4\,Hz}{\omega_{n_2}} = \frac{1}{R_2\sqrt{\underset{2\mu F}{C_C}\,\underset{1.70\mu F}{C_D}}}$$

$\boxed{R_2 = 21.6\,K\Omega}$

EXERCISE 4

In order to filter rectified EMG of a patient, a moving average FIR filters with M = 3 points is used. What is its **cut-off frequency** (-3 dB) for a sampling rate of 150 Hz?

What is the cut-off frequency if the number of moving average samples M is equal to 31?

$f_s = 150\,Hz$ and $M = 3$ $\quad f_c = ?$

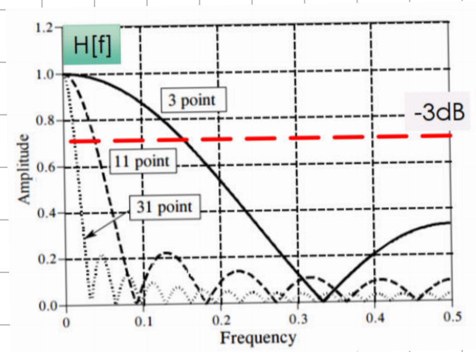
$$F_c \approx \frac{0.443}{\sqrt{M^2-1}} \cdot 100$$ (Percentage of sampling frequency)

But if instead we set a sampling frequency

$$f_c = \frac{0.443}{\sqrt{M^2-1}} \cdot f_{sampling}$$

$$f_c = \frac{0.443}{\sqrt{3^2-1}} \cdot 150\,Hz \rightarrow \boxed{23.5\,Hz}$$

$$f_c = \frac{0.443}{\sqrt{31^2-1}} \cdot 150\,Hz \rightarrow \boxed{2.15\,Hz}$$

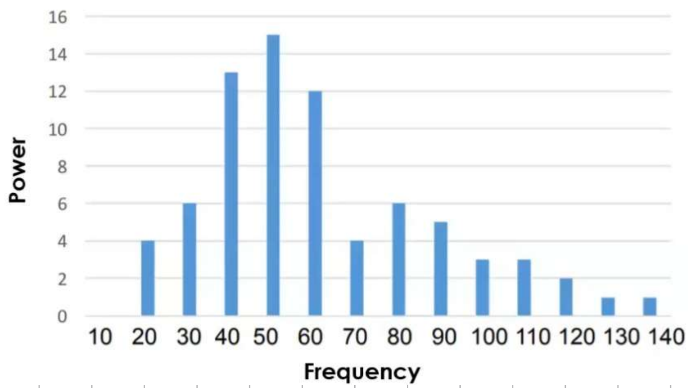


EXERCISE 6

We want to analyze the spectrum of the EMG signal extracted from the activation of different motor plates of the same motor unit. The power of the spectrum is reported below.

1.1. Calculate the Mean frequency and the Median frequency

1.2. Determine the type of symmetry and kurtosis of the spectrum.



Mean Frequency $\rightarrow \bar{f} = \frac{\sum f_i P_i}{\sum P_i} \rightarrow \bar{f} = 61.3\,Hz$

Median Frequency $\rightarrow f_m = \sum_{f_{min}}^{\bar{f}} P_i = \sum_{\bar{f}}^{f_{max}} P_i \rightarrow f_m = \frac{1}{2}\sum_{f_{min}}^{f_{max}} P_i$

$f_m = 37.5\,Hz.$

There are 75 in total

Moments

SOLUTION 1.2

Moments

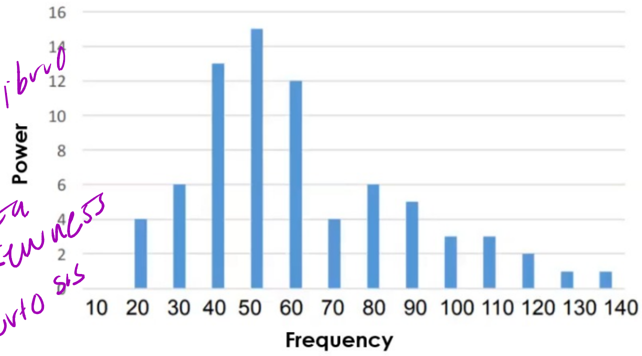
$M_0 = \sum P(f) \rightarrow M_0 = 75$ → suma de todas las probabilidades

$M_1 = \sum (f - \bar{f}) P(f) \rightarrow M_1 = 0$ → Centro de Equilibrio

$M_2 = \sum (f - \bar{f})^2 P(f) \rightarrow M_2 = 56400$ → Varianza

$M_3 = \sum (f - \bar{f})^3 P(f) \rightarrow M_3 = 1290000$ → Skewness

$M_4 = \sum (f - \bar{f})^4 P(f) \rightarrow M_4 = 132000000$ → Kurtosis



→ Always 0 if $\bar{f}$ is the average.

$(20 - 61.3)^2 \cdot 4 + \ldots$

$(20 - 61.3)^3 \cdot 4 + \ldots$

$(20 - 61.3)^4 \cdot 4 + \ldots$

SOLUTION 1.2

Variance:

$\sigma^2 = \frac{M_2}{M_0} = \frac{56400}{75} = 752$

Skewness:

$S = \frac{M_3/M_0}{\sigma^{3/2}} = \frac{1290000/75}{752^{3/2}} = 0.834$ → Positive asymmetry

Kurtosis:

$K = \frac{M_4/M_0}{\sigma^4} = \frac{132000000/75}{752^4} = 3.11$ → Leptokurtosis: more peaked than a Gaussean

Moments

$M_0 = \sum P(f) \rightarrow M_0 = 75$

$M_1 = \sum (f - \bar{f}) P(f) \rightarrow M_1 = 0$

$M_2 = \sum (f - \bar{f})^2 P(f) \rightarrow M_2 = 56400$

$M_3 = \sum (f - \bar{f})^3 P(f) \rightarrow M_3 = 1290000$

$M_4 = \sum (f - \bar{f})^4 P(f) \rightarrow M_4 = 132000000$

EXERCISE 5 – Exam Nov 5th 2019, exercise 2

An EMG signal acquisition is performed at the lower limbs of a subject during a normal walk. In Figure 1, the EMG signal of the right hamstring muscle is presented in a time window of 7 secs while in Figure 2, the power spectral density is presented.

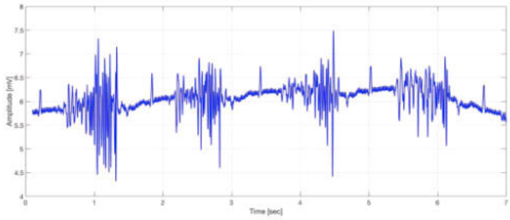


Figure 1: EMG vs time

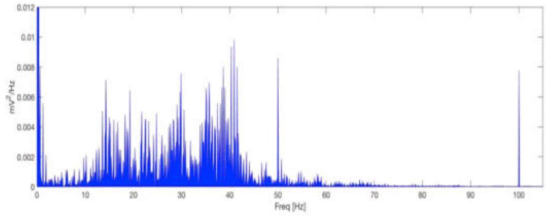


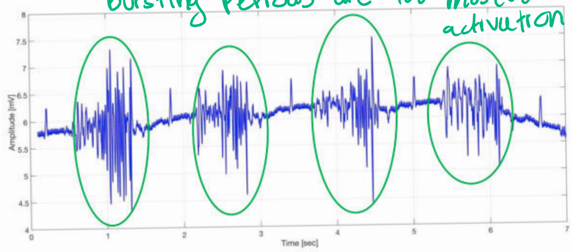
Figure 1: PSD

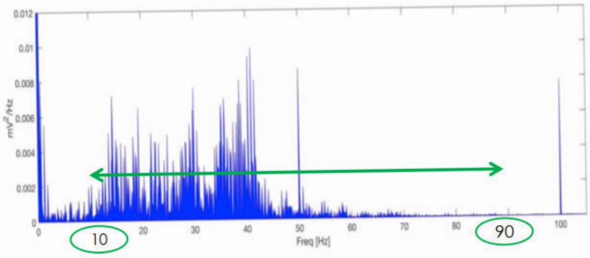
PSD: signal's power vs frequency

REQUEST AND SOLUTION

5.1. Highlight the muscle activation over time in Figure 1 and highlight the corresponding frequency components related to muscle activation in Figure 2.

Bursting Periods are the muscle activation



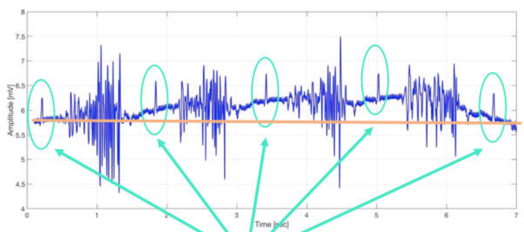


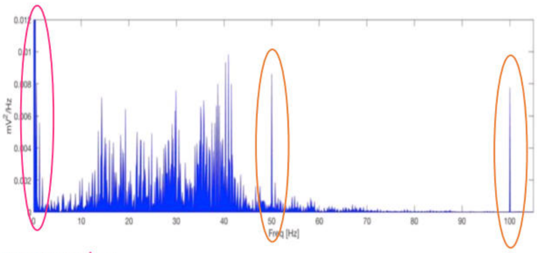
In this case Muscle activation can be commented as 10 – 90 Hz

Below 10 Hz, noises are HPF and 100 Hz there is an harmonic

REQUEST AND SOLUTION

5.2. Highlight the unwanted components, if any, in both the Figures.





Baseline drift

ECG artifacts

Low frequency noise (movement artifacts, skin-electrode interference etc.)

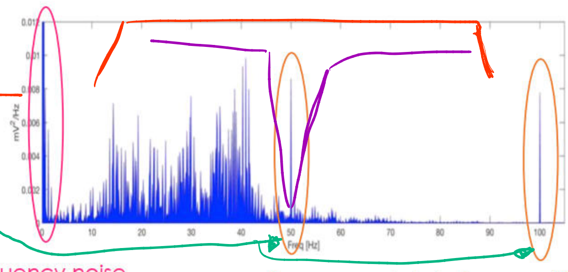
Power supply interference (50 Hz) and first harmonic (100 Hz)

REQUEST AND SOLUTION

5.3. Describe the possible solution for eliminating unwanted (noise) components in the signal.

Two examples:

1. High pass filter higher than 10 Hz + Notch filters at 50 Hz and 100 Hz
2. Band Pass Filter 10-90Hz + Notch filter at 50 Hz



Low frequency noise (movement artifacts, skin-electrode interface etc.)

Power supply interference (50 Hz) and first harmonic (100 Hz)

A post-stroke patient undergoing a neuromuscular rehabilitation protocol is monitored using surface EMG electrodes placed on the biceps brachii. The signal is sampled at 1000 Hz and is known to contain:

- Low-frequency motion artifacts due to involuntary limb tremors
- Power line interference at 50 Hz
- Baseline offset and slow drifts

Q3.1 Which filters would you apply to remove the undesired signal components? Specify their type, order, and cutoff frequencies and choose the values of the electrical components involved wether possible

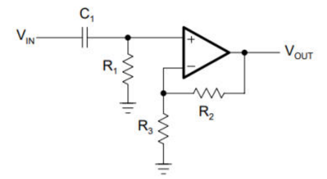
- band-pass filter of order 2 with a frequency range of 10–90 Hz
  - removes low-frequency artifacts (skin-electrode interface and motion)
  - Removes high frequency noise outside EMG signal
- notch filter centered at 50 Hz and at 100 Hz (first harmonics).

EXERCISE 6

- Design a 1st order high-pass butterworth filter with 2Hz cut-off frequency by using capacitance of 10 [μF]. What is the required resistance R?
- Design the components to have a gain of 1.9

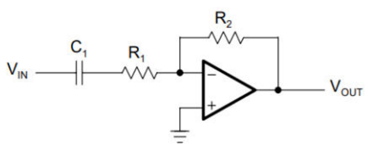
POLITECNICO
MILANO 1863

Non inverting High-Pass Filter



$$A(s) = \frac{1 + \frac{R_2}{R_3}}{1 + \frac{1}{\omega_c R_1 C_1} \cdot \frac{1}{s}}$$

First-Order Inverting High-Pass Filter



$$A(s) = -\frac{\frac{R_2}{R_1}}{1 + \frac{1}{\omega_c R_1 C_1} \cdot \frac{1}{s}}$$

Let's choose the non inverting

$$1 + \frac{R_2}{R_3} = 1.9 \longrightarrow R_2 = 9k\Omega, \quad R_3 = 10k\Omega.$$

Since $f_c = \frac{1}{2\pi R_1 C_1}$ $\quad 2Hz = \frac{1}{2\pi \cdot R_1 \cdot 10\mu F}$

$$R_1 = 7958\,\Omega$$

ASSIGNMENT (DEADLINE: TOMORROW)

POLITECNICO MILANO 1863

https://colab.research.google.com/drive/1Di2VI-Aj99qUwhbsN4XB7MLxnZ7eywww#scrollTo=-5ae86X_0h2k

1. **Signal generation:** Create a synthetic EMG signal as a combination of white Gaussian noise and bursts of activity. Add a low-frequency artifact (e.g., sinusoid at 1 Hz) to simulate movement/ECG interference. Plot the raw signal in the time and frequency domain
2. **Pre-processing:** Design and apply a Butterworth band-pass filter (20–450 Hz) to isolate the EMG band. Compare the spectrum before and after filtering.
3. **Rectification and envelope extraction:** Perform full-wave rectification. Extract the envelope using three approaches: Moving Average (e.g., 50–100 ms window), RMS in a sliding window, Low-pass Butterworth filter (e.g., 5–10 Hz cutoff) applied to the rectified signal. Plot and compare the results.
4. **Parameter estimation:** Compute the following features from the processed signal
   1. Temporal parameters: onset time, offset time, burst duration
   2. Amplitude parameters: RMS, Mean Absolute Value (MAV), Peak Amplitude
   3. Spectral parameters: Median Frequency, Mean Frequency.

   Compare the performance of the different envelope extraction methods.Comment on how the estimated parameters relate to muscle activation and signal quality.